

Instructions: Homework is graded primarily on correctness and completion.

- Write your solutions on separate sheets of paper with your name written on each problem.
 - Include all key steps and explanations that justify your answers, unless it says otherwise.
 - Submit your written work as a **.pdf** file.
 - Attach all supporting files as a **.zip** file if applicable.
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Homework Set 1

1.

Consider the rate of change equation

$$y' = \frac{y}{2}(2 + y)(y - 8),$$

which has been created to provide predictions about the future population of rabbits over time.

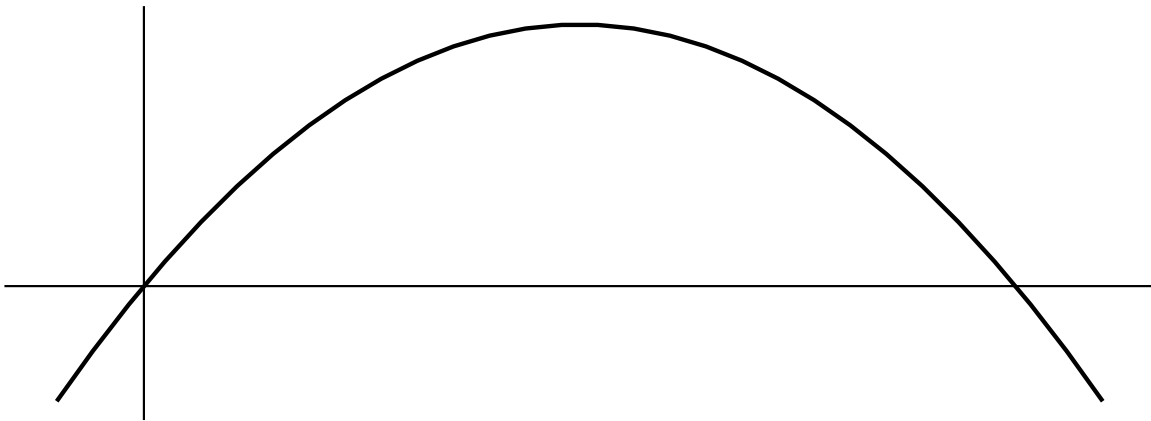
- For what values of y is $y(t)$ increasing?
- For what values of y is $y(t)$ decreasing?
- For what values of y is y' neither positive nor negative? What does this imply about the solution function $y(t)$?

2.

Valeria created the following graph to help her analyze solutions to the ODE,

$$y' = 2y \left(1 - \frac{y}{10} \right).$$

What is this a graph of (i.e., what are the axes for this graph)? What information about solutions can you infer from this graph?



3.

For the ODE $y' = 1 - y^2$,

- Sketch a phase diagram by hand and explain why you can draw such visualization for the given ODE.
- Sketch a slope field by hand and explain any shortcuts or patterns you used to make the task easier.
- Sketch several specific solutions (including the equilibriums) on the slope field.

4.

Suppose two students are memorizing the elements on a list according to the rate of change equation

$$L' = \frac{1}{2}(1 - L),$$

where L represents the fraction of the list that is memorized at time t .

- If one of the students knows one-third of the list at time $t = 0$ and the other student knows none of the list, which student is learning most rapidly at this instant? Why?
- What does the rate of change equation predict for someone who begins with the list completely memorized? Explain.
- Suppose now that the list is so long that no one can read it in a lifetime, like the decimal representation for π . In reality, no one can memorize all the digits to π , but what does the rate of change equation predict will happen for a person who starts out not knowing any of the digits? That is, according to the rate of change equation, if $L = 0$ at time $t = 0$, is there ever a value of t for which $L = 1$? Explain.

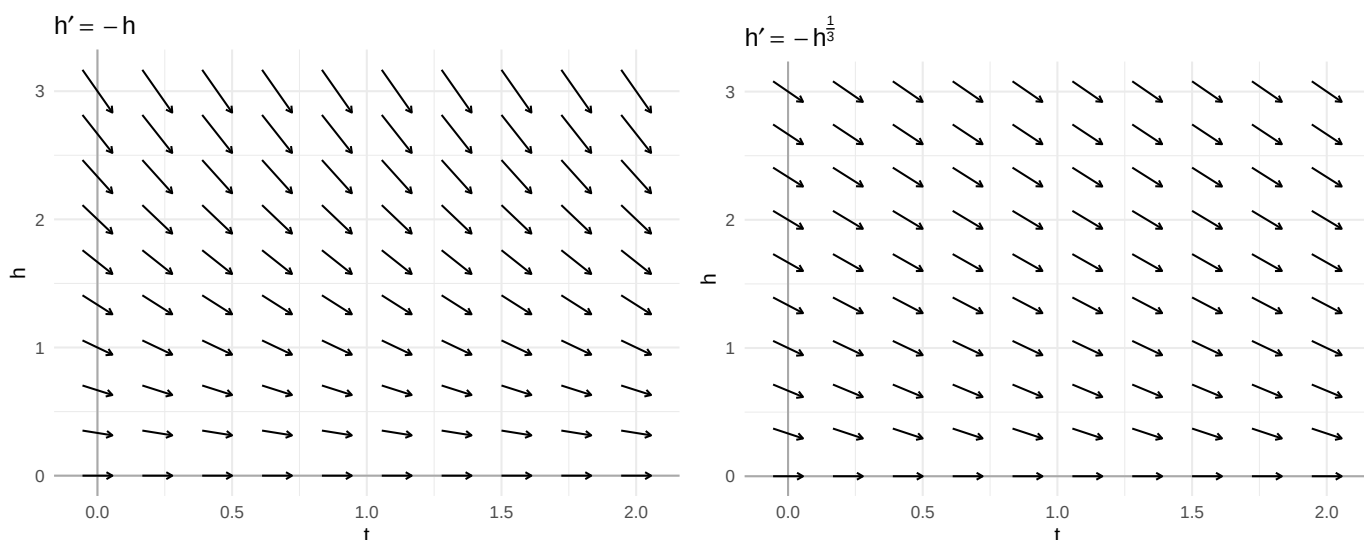
5.

A group of scientists at the Federal Aviation Association has come up with the following two different rate of change equations to predict the height of a helicopter as it nears the ground:

$$h' = -h \quad \text{and} \quad h' = -h^{\frac{1}{3}}.$$

For both rate of change equations h is in feet and t is in minutes. The scientists, of course, want their models to predict that a helicopter actually lands, but do either or both of the proposed models predict this?

- Below shows the corresponding slope fields for each ODE. Sketch in qualitatively correct graphs of the solution functions with initial conditions $h(0) = 0$ and $h(0) = 2$.



- For each differential equation, use the uniqueness theorem to show whether the graphs of each ODE will ever touch or cross the equilibrium. Interpret the results in context.

6.

Consider the ODE with initial condition,

$$y' + ty = e^{-\frac{1}{2}t^2}, \quad y(0) = 1.$$

Show that $y(t) = te^{-\frac{1}{2}t^2} + e^{-\frac{1}{2}t^2}$ is a solution and explain why this is a specific solution.

7.

For each ODE,

- determine all equilibrium solutions if any, and
- use the existence and uniqueness theorem to determine whether or not graphs of solution functions will ever touch any and all equilibrium solutions.

a) $y' = 0.5(1 - y)$ b) $y' = 0.3y \left(1 - \frac{y}{10}\right)$ c) $y' = -t + 1$ d) $y' = y^{\frac{1}{2}}$

8.

The following questions relates to the general definition of ODEs and their solutions. Answer each question with at least two to three sentences.

- In algebra, the goal of solving an equation such as $x^2 + 4x = 2$ is to find the values of x that make a true statement. In differential equations, what is the goal of solving an equation such as $x' + 4x = 2$? Explain your reasoning.
- The letter y appears in two places in the differential equation $y' = 0.3y$. Is it appropriate to think of both occurrences of y as function of t ? Explain your reasoning.
- What are the key differences between general and specific solutions of an ODE? Explain your reasoning.
- An equilibrium solution is a constant solution where the rate of change is zero in the long term. Is it true that equilibrium solutions are nullclines but not all nullclines are equilibriums? Explain your reasoning.

9.

Construct one ODE for each of the following classifications. For each case, briefly explain why the equation satisfies the specified criteria or justify why such an equation cannot exist.

- 1st-order, homogeneous, non-autonomous, linear
- 1st-order, non-homogeneous, autonomous, non-linear
- 1st-order, non-homogeneous, non-autonomous, linear
- 2nd-order, homogeneous, non-autonomous, linear
- 2nd-order, non-homogeneous, autonomous, non-linear
- 2nd-order, non-homogeneous, non-autonomous, linear

10. Challenge Problem

Consider the 1st-order ODE,

$$y' = n(1 - y) - (1 - t^{n-1})$$

where n is a positive integer, $n \geq 0$.

- a) Classify the given differential equation in terms of order, autonomy, homogeneity, and linearity for cases when $n = 0$, $n = 1$, and $n > 1$.
- b) Determine the values of n where there are equilibrium solutions and values of n where there are none.
- c) Create a function for the isoclines and nullclines for each case of n , and draw these functions on your slope fields.
- d) Draw slope fields for cases when $n = 0$, $n = 1$, and $n > 1$ (for even and odd numbers, give one example each), and then draw at least four qualitatively different solution curves (including equilibria if any).